

CRM Project Overview

This document presents the rationale, requirements, intended uses, and complete setup instructions for the Spring Boot-based Customer Relationship Management (CRM) application.

1. Necessity of the CRM Project

- **Centralized Customer Data:** Organizations require a single source of truth for customer information, interactions, and transaction history.
 - **Process Automation:** Manual handling of customer follow-ups, leads, and support requests is error-prone and inefficient.
 - **Scalability:** As businesses grow, tracking hundreds or thousands of client records demands a scalable software solution.
 - **Improved Decision-Making:** Real-time dashboards and reports enable informed decisions on sales strategies and resource allocation.
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2. Project Requirements

2.1 Functional Requirements

- **User Authentication & Authorization:** Secure login, registration, and role-based access control.
- **Customer Management:** Create, read, update, and delete (CRUD) operations for customer entities.
- **Product & Offer Management:** CRUD operations for products, categories, and special offers.
- **Meeting & Interaction Records:** Schedule and log customer meetings, calls, and follow-ups.
- **Search & Filtering:** Search customers by name, email, or phone.
- **API Documentation:** Interactive REST API documentation via Swagger UI.

2.2 Non-Functional Requirements

- **Performance:** Fast response times with pagination for large datasets.
 - **Security:** JWT-based authentication, input validation, and password encryption.
 - **Maintainability:** Clear modular architecture (controllers, services, repositories, DTOs).
 - **Scalability:** PostgreSQL database with Hibernate for efficient data management.
 - **Usability:** Intuitive API endpoints documented in Swagger.
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3. Intended Uses

- **Sales Teams:** Track leads, opportunities, and customer interactions.
 - **Support Departments:** Log and manage customer support tickets and follow-ups.
 - **Marketing Analysts:** Generate reports on customer segments and campaign effectiveness.
 - **Management:** Monitor key performance indicators (KPIs) and pipeline health.
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4. Setup Instructions

4.1 PostgreSQL Database Setup

Install PostgreSQL (skip if already installed)

Download from: <https://www.postgresql.org/download/>

Open PostgreSQL shell as superuser

```
psql -U postgres
```

Create the CRM database

```
CREATE DATABASE crm;
```

Create a dedicated database user

```
CREATE USER crm_user WITH PASSWORD 'crm_pass';
```

Grant privileges to the user

```
GRANT ALL PRIVILEGES ON DATABASE crm TO crm_user;
```

Connect to the crm database

```
\c crm
```

Grant schema permissions

```
GRANT USAGE ON SCHEMA public TO crm_user;
```

```
GRANT CREATE ON SCHEMA public TO crm_user;
```

Optional: Grant privileges on existing tables and sequences

```
GRANT SELECT, INSERT, UPDATE, DELETE ON ALL TABLES IN SCHEMA public TO crm_user;
```

```
GRANT USAGE, SELECT ON ALL SEQUENCES IN SCHEMA public TO crm_user;
```

```
# Ensure default privileges for future objects
```

```
ALTER DEFAULT PRIVILEGES IN SCHEMA public GRANT SELECT, INSERT, UPDATE, DELETE ON  
TABLES TO crm_user;
```

```
ALTER DEFAULT PRIVILEGES IN SCHEMA public GRANT USAGE, SELECT ON SEQUENCES TO  
crm_user;
```

4.2 Application Project Setup

```
# 1. Download the project ZIP from:
```

```
# https://github.com/nidabaser/CRM\_Spring\_Boot/archive/refs/heads/main.zip
```

```
# 2. Extract the ZIP archive
```

```
cd path/to/CRM_Spring_Boot-main
```

```
# 3. Verify Java 17 is installed
```

```
java -version # Should report Java 17.x
```

```
# 4. Set JAVA_HOME (Windows PowerShell as Administrator)
```

```
setx JAVA_HOME "C:\Program Files\Java\jdk-17" /M
```

```
setx PATH "$env:JAVA_HOME\bin;$env:PATH" /M
```

```
# Restart terminal and verify
```

```
java -version
```

```
# 5. Import into IDE
```

```
# - IntelliJ IDEA: Open Project > select extracted folder
```

```
# - VS Code: Open Folder > ensure Maven extension is active
```

```
# 6. Build dependencies and compile
```

```
./mvnw clean install # Linux/macOS
```

```
mvnw.cmd clean install # Windows
```

7. Run the application

```
./mvnw spring-boot:run
```

or

```
java -jar target/crm-0.0.1-SNAPSHOT.jar
```

8. Access the application

Swagger UI: <http://localhost:8080/swagger-ui/index.html>

5. Verification and Validation

- Use Swagger UI to verify all endpoints are functional.
 - Create sample customers, products, and meeting records.
 - Check database tables in pgAdmin for proper data persistence.
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